

# Emma Dann

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## Education

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**PhD in Biological Sciences (Computational Biology)** – University of Cambridge, UK (Oct 2019 – Oct 2023)

Advisors: Prof. Sarah Teichmann, Dr. John Marioni

Thesis title: *Discovering variation from cell atlases: comparative methods for single-cell genomics*

**MSc in Cancer, Stem Cells and Developmental Biology (Bioinformatics profile)** – Utrecht University, NL (Sep 2017 – Aug 2019)

**BSc in Biomolecular Sciences and Technology** – University of Trento, IT (Sep 2014 – Jul 2017)

## Research experience

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Post-doctoral fellow – **Stanford University, Palo Alto, US** (May 2024 – present)

Visiting post-doctoral fellow – **Gladstone Institute of Genomic Immunology, San Francisco, US**

Advisors: Prof. Jonathan Pritchard, Prof. Alex Marson

Graduate researcher – **Wellcome Sanger Institute, Cambridge, UK** (Oct 2019 – Jan 2024)

Advisors: Prof. Sarah Teichmann, Dr. John Marioni

Research intern – **European Molecular Biology Laboratory, Heidelberg, DE** (Sep 2018 – Aug 2019)

Advisors: Dr. Wolfgang Huber

Research intern – **KNAW Hubrecht Institute, Utrecht, NL** (Sep 2017 – Aug 2018)

Advisors: Prof. Alexander van Oudenaarden

Research intern – **Center for Integrative Biology, University of Trento, IT** (Feb 2017 – Jul 2017)

Advisors: Prof. Francesca Demichelis

## Fellowships & awards

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### Fellowships

**Helen Hay Whitney fellowship** (Nov 2024 – present)

Postdoctoral fellowship, awarded \$229,500 over 3 years

**European Molecular Biology Organization (EMBO) long-term fellowship** (Jul 2024 - present)

Postdoctoral fellowship, awarded €130,800 over 2 years

**Wellcome Sanger Institute PhD studentship**, Wellcome Trust, UK (Oct 2019 - Oct 2023)

PhD fellowship, awarded £100,000 plus tuition over 4 years

**U/Select honours programme** Utrecht University, NL (Jan 2018 - Oct 2019)

Master's honors program, awarded €2,000 travel grant for international research internship

**Merit Award** University of Trento, IT (Oct 2017)

Prize for outstanding graduates, awarded €3,000

### Funding

**Anthropic AI for Science program** (2026)

Awarded \$20,000 over 6 months

**Contributed as key personnel to NIH R01 grant (2R01HG008140) (2025)**

PIs: Jonathan K. Pritchard, Alex Marson

Reviewed as 1st percentile score

**Contributed as key personnel to ERC Advanced Grant (ThyDESIGN) (2022)**

PI: Sarah Teichmann

Awarded €2.5M over 5 years

## Other awards

**Tahoe Deep Dive Hackathon, first prize winner** (May 2025)

Awarded \$20,000 for best project on prediction of clinical outcomes from large-scale drug perturbation datasets.

## Publications

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► denotes manuscripts as leading author, \* denotes equal contribution

### Pre-print Manuscripts

Hartman A, Blair J., ... , **Dann E.**, ... , Satpathy A., Roth T. (2026) Systematic identification of seed-driven off-target effects in Perturb-seq experiments. *bioRxiv* | [Full text](#)

► Zhu R.\*, **Dann E.\***, Yan J., Retana J.R., Goto R., Guitche R., Brixi L., ... , Pritchard J.K., Marson A. (2025) Genome-scale perturb-seq in primary human CD4+ T cells maps context-specific regulators of T cell programs and human immune traits. *bioRxiv (accepted at Cell)* | [Full text](#) | [code](#) | [data](#)

Pett J.P., Prete M., Pham D., England N., ... , **Dann E.**, ... , Teichmann S.A., Meyer K. (2025) An atlas of TF driven gene programs across human cells. *bioRxiv* | [Full text](#)

► **Dann E.\***, Teeple E.\*, Elmentaite R., ... , Rinaldis E., Teichmann S.A. (2024) Estimating the impact of single-cell RNA sequencing of human tissues on drug target validation., *medRxiv* | [Full text](#) | [code](#)

### Peer-reviewed Manuscripts

Rathore U., Dugan E., Thornton H., ... , **Dann E.**, ... , Marson A. (2026) Systematic discovery of pro- and anti-HIV host factors in primary human CD4+ T cells. *Cell* | [Full text](#)

Deconinck L, Zappia L, Cannoodt R, Morgan M, **scverse core**, ... , Saeys Y. (2025) *anndataR* improves interoperability between R and Python in single-cell transcriptomics. *Bioinformatics* | [Full text](#)

Heumos L., Ji Y., ... , **Dann E.**, ... , Theis F.J. (2025) Pertpy: an end-to-end framework for perturbation analysis. *Nature Methods* | [Full text](#) | [python package](#)

Ota M., Spence J.P., Zeng T., **Dann E.**, Milind N., Marson A., Pritchard J.K. (2025) Causal modeling of gene effects from regulators to programs to traits: integration of genetic associations and Perturb-seq. *Nature* | [Full text](#)

Luecken M., Gigante S., Burkhardt D., Cannoodt R., ... , **Dann E.**, ... , Saeys Y., Theis F.J., Krishnaswamy S. (2025) Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology* | [Full text](#)

Schuster V., **Dann E.**, Krogh A., Teichmann S.A. (2024) multiDGD: A versatile deep generative model for multi-omics data. *Nature Communications* | [Full text](#) | [python package](#)

Yayon N., Kedlian V., Boehme L., ... , **Dann E.**, ... , Uhlmann V., Notarangelo L.D., Germain R.N., Radtke A.J., Marioni J.C., Taghon T., Teichmann S.A. (2024) A spatial human thymus cell atlas mapped to a continuous tissue axis. *Nature* | [Full text](#)

Oliver A.J., Ni H., ... , **Dann E.**, ... , Elmentaite R., Teichmann S.A. (2024) Single-cell integration reveals metaplasia in inflammatory gut diseases. *Nature* | [Full text](#)

Missarova A., **Dann E.**, ... , Marioni J.C. (2023) Leveraging neighborhood representations of single-cell data to achieve sensitive DE testing with miloDE. *Genome Biology* | [Full text](#) | [R package](#)

Sumanaweera D., Suo C., Muraro D., **Dann E.**, ... , Teichmann S.A. (2023) Gene-level alignment of single cell trajectories. *Nature Methods* | [Full text](#) | [python package](#)

- Barnes J. L., He P., ... , **Dann E.**, ... , Teichmann S.A., Meyer K. B., Nikolic M. Z. (2023) Early human lung immune cell development and its role in epithelial cell fate. *Science Immunology* | [Full text](#)
- **Dann E.**, Cujba A.M., Oliver A., Meyer K., Teichmann S.A., Marioni J.C. (2023) Precise identification of cell states altered in disease using healthy single-cell references, *Nature Genetics* | [Full text](#) | [code](#) | [python package](#)
- Botting R.A., Goh I., ... , **Dann E.**, ... , Teichmann S.A. , Haniffa M. (2023) Yolk sac cell atlas reveals multiorgan functions during human early development. *Science* | [Full text](#)
- Suo C., Polanski K., **Dann E.**, ... , Tuong Z.K., Clatworthy M., Teichmann S.A. (2023) Dandelion uses the single-cell adaptive immune receptor repertoire to explore lymphocyte developmental origins, *Nature Biotechnology* | [Full text](#) | [python package](#)
- Heumos L., ... **Single-cell Best Practices Consortium**, ... , Theis F.J. (2023) Best practices for single-cell analysis across modalities. *Nature Reviews Genetics* | [Full text](#)
- Suo C.\*, **Dann E.\***, ... , Haniffa M., Teichmann S.A. (2022) Mapping the developing human immune system across organs. *Science* | [Full text](#) | [code](#) | [data](#)
- Lalchand V.\*, Ravuri A.\*, **Dann E.\***, ..., Madad S., Teichmann S.A., Lawrence N.D. (2022) Modelling Technical and Biological Effects in scRNA-seq data with Scalable GPLVMs. *Proceedings of the 17th Machine Learning in Computational Biology meeting* | [Full text](#)
- He P., Lim K., Sun D., ... , **Dann E.**, ... , Meyer K.B., Rawlins E.L. (2022) A human fetal lung cell atlas uncovers proximal-distal gradients of differentiation and key regulators of epithelial fates, *Cell* | [Full text](#)
- Kleshchevnikov V., Shmatko A., **Dann E.**, ..., Stegle O., Bayraktar O.A. (2022) Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature Biotechnology* | [Full text](#) | [python package](#)
- **Dann E.**, Henderson N.C., Teichmann S.A., Morgan M.D., Marioni J.C. (2022) Differential abundance testing on single-cell data using k-nearest neighbor graphs, *Nature Biotechnology* | [Full text](#) | [R package](#) | [python package](#)
- Jardine L., Webb S., ... , **Dann E.**, ... , Göttgens B., Roberts I., Teichmann S. and Haniffa M. (2021) Blood and immune development in human fetal bone marrow and Down syndrome, *Nature* | [Full text](#)
- Elmentaite R., Kumasaka N., Roberts K., Fleming A., **Dann E.**, ... , James K.R., Teichmann S.A. (2021), Cells of the human intestinal tract mapped across space and time, *Nature* | [Full text](#)
- Stephenson E., Reynolds G., Botting R.A., Calero-Nieto F.J., Morgan M.D., Tuong Z.K., Bach K., Sungnak W., ... , **Dann E.**, ... , Duncan C.J.A, Smith K., Teichmann S.A., Clatworthy M.R., Marioni J.C., Gottgens B., Haniffa M. (2020) Single-cell multi-omics analysis of the immune response in COVID-19, *Nature Medicine* | [Full text](#)
- Beltran H., Romanel A., ... , **Dann E.**, ... , Benelli M., and Demichelis F. (2020) Circulating Tumor DNA Profile Recognizes Transformation to Castration-Resistant Neuroendocrine Prostate Cancer, *Journal of Clinical Investigation* | [Full text](#)

## Presentations

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### Conference oral presentations

- June 2026 - Biology at Scale conference, Wellcome Genome Campus, Hinxton (UK)
- May 2026 - Warren Alpert Computational Biology and AI Research Symposium, Stanford (US)
- Feb 2026 - Arc Institute Symposium, San Francisco (US)
- Apr 2025 - Chan Zuckerberg Initiative Cell Science Meeting, Chicago (US)
- Nov 2022 - Machine Learning in Computational Biology conference, virtual
- Jun 2022 - Single Cell Biology conference, Wellcome Genome Campus, Hinxton (UK)
- May 2022 - The Biology of Genomes, Cold Spring Harbor Laboratory (US)
- Oct 2021 - Chan Zuckerberg Initiative Single Cell Biology Meeting, virtual
- Sep 2021 - Next Generation Genomics Meeting, virtual
- Mar 2021 - CZI Seed Networks Computational biology meeting, virtual

## Invited research seminars

Apr 2026 - Emerging Genomic Scientists Symposium, University of Pennsylvania, Philadelphia (US)  
Oct 2025 - Ultima Genomics user group session @ American Society for Human Genetics, Boston (US)  
Sep 2025 - scverse community meeting, virtual  
Jul 2025 - AI + Biomedicine seminar, Stanford (US)  
Sep 2024 - BRAID Genentech, South San Francisco (US)  
Sep 2024 - Human Genetics department, University of Chicago, Chicago (US)  
Sep 2024 - SQLIFTS Lecture Series, Northwestern University, Chicago (US)  
Mar 2024 - Human Technopole, Milan (IT)  
Dec 2023 - Google DeepMind, London (UK)  
Jul 2023 - Sanofi Precision Medicine & Computational biology, Cambridge (US)  
Jun 2023 - Chan Zuckerberg Initiative, Redwood City (US)  
Apr 2023 - Cambridge AI club for biomedicine, Milner Institute, Cambridge (UK)  
Dec 2022 - UMC Utrecht single-cell genomics meeting, virtual  
Sep 2022 - UK Conference of Bioinformatics and Computational Biology, virtual  
Jul 2022 - Computational Health Center seminar, Helmholtz Munich (DE)  
Apr 2022 - Maxwell Society Annual Conference, King's College London (UK)  
Apr 2022 - NYU Langone single cell journal club, virtual  
Mar 2021 - Cambridge Center for Physical Biology Single Cell Symposium, virtual

## Guest lectures

Aug 2024 - Why there's never been a better time to study human biology, [Girls in Research](#) program (virtual)  
Aug 2023 - Intro to single-cell genomics series, University of Sri Jayewardenepura (virtual)  
Mar 2023 - Introduction to multiomics data integration and visualisation, EMBL-EBI training course (UK)  
Apr 2022 - Best practices and open challenges in single-cell genomics data analysis, Maxwell Society Annual Conference, King's College London (UK)

## Conference poster presentations

Oct 2025 - American Society for Human Genetics conference, Boston (US)  
Jul 2023 - Human Cell Atlas General Meeting, Toronto (CA)  
Oct 2022 - Single Cell Genomics conference, Utrecht (NL)  
Jun 2021 - Human Cell Atlas General Meeting, virtual

## Teaching & mentoring

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**Computational research assistant recruitment and mentoring, Marson lab** (Fall 2025 - present)

**PhD rotation student recruitment and mentoring, Pritchard lab** (Winter 2025 - present)

**Stanford & Tanzania Human Genetics Organization virtual training program** (Spring 2025)

One-on-one virtual mentoring program on research project for students from Tanzania

**PhD rotation student recruitment and mentoring, Teichmann lab** (Fall 2021 - Fall 2023)

**Systems biology: From large datasets to biological insight, EMBL-EBI training course** (Jul 2022)

Invited instructor for single-cell multi-omics module ([course material](#))

**Advanced topics in Single Cell Genomics, Swiss Institute for Bioinformatics course** (Apr 2022)

Invited instructor for multi-omics module ([course material](#))

**Advanced topics in Single Cell Genomics, SciLifeLab / Swiss Institute for Bioinformatics Summer School** (Aug 2021)

Invited instructor for multi-omics module ([course material](#))

# Leadership & outreach

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## **Core team member, [scverse consortium](#)** (Feb 2024 – present)

I am part of the governance team of the consortium, overseeing the development and maintenance of foundational software tools for omics data analysis.

- Organizational governance and strategic planning
- Development of academic and industry partnerships for funding initiatives and collaborative research projects
- Lead organizer of the [scverse annual conference](#) and [community hackathons](#)
- Contributor to open-source [tutorials](#) and [resources](#) for newcomers to single-cell genomics analysis in python

## **Competition judge, [Open Problems Single-cell perturbations Kaggle competition](#)** (Fall 2023)

Served as judge for NeurIPS-affiliated machine learning challenge focused on predicting drug perturbation responses.

## **Co-organizer, [Strengthening integration pipelines session, Human Cell Atlas general meeting](#)** (Jul 2023)

Co-organized and led panel discussion on cell atlas construction and usage.

## **Member of the selection panel, [Sanger prize competition](#)** (Mar 2023)

The Sanger Prize competition funds research internships at Sanger for undergraduate students from LMI countries. I was a member of the Grad Panel, evaluating essays from applicants.

## **Co-organizer [Open Problems in Single-Cell Analysis Jamboree](#)** (April 2021)

Led task for benchmarking of compositional analysis methods for scRNA-seq.

## **Co-organizer, [EBI-Sanger-Cambridge PhD Symposium](#)** (Feb 2020)

I was in charge of the scientific program and sponsorship for the PhD program symposium.

## **Board member, [Open Wet Lab \(OWL\)](#)** (Oct 2015 – Jan 2018)

OWL is the first biohacking organization in Italy. I was in charge of the association's budget and directed the publication of OWL's blog and scientific outreach articles for local newspapers.

# Peer review

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**Journals:** Nature (1 paper), Cell (1\* paper), Nature Biotechnology (2\* papers), Nature Genetics (1 paper), Nature Communications (1 paper), Genome Biology (1 paper), Bioinformatics Advances (1 paper), OUP Bioinformatics (1 paper), Reviewer Commons (1 paper) \*Assisting a senior reviewer

**Conferences:** Machine Learning in Computational Biology (2023, 2024, 2026)

**Grants/fellowships:** Chan Zuckerberg Initiative Single Cell Biology Data Insights, Christian Doppler Research Association Transfer.S2S grants

# Software

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**pertpy** |  [scverse/pertpy](#) | contributor | perturbation analysis toolkit for the scverse ecosystem.

**miloR** |  [MarioniLab/miloR](#) | primary developer | R/Bioconductor package for the identification of differential cell abundance in KNN graphs for single-cell data

**milopy** |  [emdann/milopy](#) | primary developer | python package for the identification of differential cell abundance in KNN graphs for single-cell data

**genomic-features** |  [scverse/genomic-features](#) | primary developer | Genomic annotations using Bioconductor resources in Python.

**oor\_benchmark** |  [MarioniLab/oor\\_benchmark](#) | primary developer | python framework to benchmark workflows for detection of out-of-reference cells in comparative single-cell analysis

**cell2location** |  [BayraktarLab/cell2location](#) | contributor | python package for mapping of tissue cell types via integration of single cell RNA-seq and spatial transcriptomics data

## Industry experience

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**Data science consultant – Ensocell Ltd** (Oct 2023 – Mar 2024)

Target discovery platform and analysis of single-cell data